Bayesian circuit learning at mean-field scaling: a variational principle and width-independent generalization

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Abstract

We prove polynomial sample complexity for Bayesian learning of Boolean functions by fixed-depth tanh networks at mean-field scaling. The Gaussian prior has first-layer standard deviation $d^{-1/2}$, internal-layer standard deviation $N^{-1/2}$, and readout standard deviation N^{-1}. It is first restricted to networks with absolute margin greater than $1/2$ on the entire Boolean cube; learning then conditions on noiseless labels. For every sufficiently large width, a sample bound polynomial in the input dimension and the target's bounded-weight network complexity ensures correct-label posterior probability strictly greater than $2/3$ on almost all test inputs. The sample bound is independent of width; the required width threshold is not quantified. The proof identifies the exact finite-domain large-deviation rate, shows that full-margin minimizers have quadratic feature tilts with polynomially bounded entrywise coefficients, and realizes their feature covariances by polynomial finite networks. We also prove a target-mass bound for every fixed depth and all $\gamma \geq 1$ in the strong-prior family, and a shallow-network learning theorem valid at every width throughout that family.

1 Model and conclusions

Let $d \geq 1$, $L \geq 2$, $\mathcal{X} = \{-1, 1\}^d$, and $\bar{x} = (1, x_1, \ldots, x_d)$. For a bounded 1-Lipschitz activation $\phi : \mathbb{R} \to [-1, 1]$, a depth-$L$, hidden-width-$N$ network is

$$h^0(x) = \bar{x}, \qquad h^s(x) = \phi(W^s h^{s-1}(x)) \ (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no independent hidden biases beyond the constant input coordinate. For a Boolean function f define

$$A_{N,f} = \{w : f(x) z_w(x) > 1/2 \text{ for all } x \in \mathcal{X}\}, \qquad A_N = \bigcup_f A_{N,f}.$$

Let $C_L(f)$ be the least integer $C \geq 1$ for which a width-C network implements f with strict signed output margin $1/2$ on the entire Boolean cube and all its weights bounded in absolute value by C. We assume this complexity is finite for the target.

Let $\mu_{N,\gamma}$ be the product Gaussian law with standard deviations

$$\sigma_1 = N^{(1-\gamma)/2}/\sqrt{d}, \quad \sigma_s = N^{-\gamma/2} \ (2 \leq s < L), \quad \sigma_L = N^{-(\gamma+1)/2}.$$

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Define the weighted norm and the induced Boolean prior by

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1}\|W^s\|_F^2 + N\|W^L\|_2^2, \qquad \pi_{N,\gamma}(f) = \frac{\mu_{N,\gamma}(A_{N,f})}{\mu_{N,\gamma}(A_N)}.$$

For $x_1, \dots, x_n$ drawn independently from any distribution D on $\mathcal{X}$, the posterior is this prior conditioned on $g(x_i) = f(x_i)$ for all i. Write

$$R_D(\Pi) = \mathbb{E}_{g\sim\Pi}\mathbb{P}_{x\sim D}\{g(x) \neq f(x)\}.$$

The normalization by $\mu_{N,\gamma}(A_N)$ cancels from every prior ratio used below. The global margin restriction is essential to this model description; we do not replace it by training-set margins.

Theorem 1.1 (Mean-field learning at sufficiently large width)**.** *Fix depth $L \geq 2$ and activation $\phi = \tanh$. There is a polynomial p_L with the following property. For every target f with $C = C_L(f) < \infty$ and every $\varepsilon \in (0,1)$, there is a width threshold $N_0 = N_0(d, L, f, \varepsilon) \geq C$ such that for every $N \geq N_0$, every $\delta \in (0,1)$, every input distribution D, and every*

$$n \geq \frac{6}{\varepsilon}\left[p_L(d, C) + \log(1/\delta)\right],$$

with probability at least $1-\delta$ over the training sample S_n,

$$R_D(\Pi_{N,1}) \leq \varepsilon/3, \qquad \mathbb{P}_{x\sim D}\{\Pi_{N,1}(g(x) = f(x) \mid S_n) > 2/3\} \geq 1-\varepsilon.$$

Here fixed depth means that both the coefficients and degree of p_L may depend on L. The polynomial is independent of width. No polynomial bound on N_0, or all-width version of this deep-network statement, is asserted.

Theorem 1.2 (Uniform shallow-network learning)**.** *For $L = 2$ there is a polynomial p such that the following holds for every $\gamma \geq 1$, every target f with $C = C_2(f)$, every $N \geq C$, every $\varepsilon, \delta \in (0,1)$, and every input distribution D. If*

$$n \geq \frac{2}{\varepsilon}\left[p(d, C, \log(2/\varepsilon)) + \log(1/\delta)\right],$$

then, with probability at least $1-\delta$ over the training sample, $R_D(\Pi_{N,\gamma}) \leq \varepsilon$. The same posterior-majority classifier has error at most 2ε. The polynomial is independent of both N and γ.

Corollary 1.3 (Confident correct prediction)**.** *Under the hypotheses of Theorem 1.2, if*

$$n \geq \frac{6}{\varepsilon}\left[p(d, C, \log(6/\varepsilon)) + \log(1/\delta)\right],$$

then with training-sample probability at least $1-\delta$,

$$\mathbb{P}_{x\sim D}\left\{\Pi_{N,\gamma}(g(x) = f(x) \mid S_n) > \frac{2}{3}\right\} \geq 1-\varepsilon.$$

Thus an input counts as successful only when the posterior assigns its correct label probability strictly greater than $2/3$.

Proof. On every input failing the displayed confidence condition, the posterior error probability is at least $1/3$. Consequently the probability of confidence failure is at most $3R_D(\Pi)$. Apply Theorem 1.2 with risk tolerance $\varepsilon/3$. □

The first theorem treats the infinite-width mean-field regime through a statement valid for all sufficiently large finite widths. The second theorem is stronger in its width quantifier for shallow networks and requires only a bounded Lipschitz activation. The proof first establishes a common target-mass bound. An exact large-deviation representation then identifies the critical rate, and a new forward construction compresses its full-margin minimizers. A finite relevant-function argument converts the resulting count into the learning guarantee. The separate shallow argument gives its all-width result without a limiting variational analysis.

2 A cylinder construction valid throughout $\gamma \geq 1$

The following construction bounds the target's prior cost at every width $N \geq C$, without a width threshold diverging as $\gamma \downarrow 1$.

Proposition 2.1 (Uniform cloning mass bound)**.** *Fix $L \geq 2$ and a bounded 1-Lipschitz activation. There is a function $Q_L(d, C)$ bounded by a polynomial such that for every $\gamma \geq 1$, every Boolean target with $C_L(f) \leq C$, and every $N \geq C$,*

$$-\frac{2}{N^\gamma} \log \Pr\{f(x) z_w(x) > 1/2 \text{ for all } x\} \leq Q_L(d, C).$$

The prior variances and absence of later-layer biases are exactly those in the main model.

Proof. Choose a width-C teacher, increase its readout by a factor of three, and write its hidden matrices as B^s and readout as v. Thus the teacher has margin strictly greater than $3/2$, hidden entries bounded by C, and readout entries bounded by $3C$. If the original witness has smaller width, first use its actual width in the construction below; all estimates only improve. Partition $\{1, \dots, N\}$ into C groups of sizes k_a differing by at most one. Then

$$1 \leq k_a, \qquad \frac{\max k_a}{\min k_a} \leq 2, \qquad \frac{N}{k_a} \leq 2C.$$

Let $u_a = k_a^{-1/2} \mathbf{1}_{\text{group } a}$ and let P be the orthogonal projection onto their span. The exact clone w_* has first-layer row B_b^1 for a neuron in group b, hidden entry B_{ba}^s / k_a from group a to group b, and readout v_b / k_b on group b. Every hidden activation of the exact clone is group-constant. Moreover,

$$\|W_*^s\|_{\text{op}} \leq \sqrt{2} C^2 \quad (2 \leq s < L), \qquad \sqrt{N} \|W_*^L\|_2 \leq 3\sqrt{2} C^2.$$

The first bound follows by restricting to the group-constant subspace: the resulting matrix has entries $\sqrt{k_b / k_a} B_{ba}^s$, and its Frobenius norm is at most $\sqrt{2} C^2$.

For fixed $\gamma \geq 1$, set $t = t(N) = N^{(1-\gamma)/2} \leq 1$. This is the prescribed width-dependent Gaussian prefactor; it is constant in N only at $\gamma = 1$. The random coordinates $\sqrt{d} W_{ij}^1$, $\sqrt{N} W_{ij}^s$ for $2 \leq s < L$, and $\alpha_j = N W_j^L$ are independent centered Gaussians of standard deviation t. Impose the following cylinder event $\mathcal{C}_\rho$, for $0 < \rho \leq 1$:

- every $\sqrt{d} W_{ij}^1$ is within ρt of $\sqrt{d} (W_*^1)_{ij}$;
- for every hidden row j and group a, the coordinate $\langle \sqrt{N} W_{j,\cdot}^s, u_a \rangle$ is within ρt of its cloned value $\sqrt{N / k_a} B_{ba}^s$;
- every α_j is within ρt of its cloned value $N v_b / k_b$.

All these constraints concern independent scalar Gaussian coordinates. The orthogonal coordinates of each internal row remain independent prior Gaussians. Consequently write

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad E^s P = 0.$$

On the cylinder, $\|R^s\|_{\mathrm{op}} \le \|R^s\|_F \le \rho t\sqrt{C}$. The free matrix is distributed as $(t/\sqrt{N})G^s(I-P)$, with G^s a standard Gaussian matrix, independently of the cylinder coordinates. Let

$$b_L = 2 + \sqrt{2\log(2L)}.$$

The Gaussian operator-norm bound and a union bound imply, conditionally on $\mathcal{C}_\rho$, that all $\|E^s\|_{\mathrm{op}} \le b_L t$ with probability at least $1/2$. This is also valid when $L = 2$, where there are no such matrices.

Let $e_s(x) = N^{-1/2}\|h^s_w(x) - h^s_*(x)\|_2$. The input constraints give $e_1(x) \le \rho(d+1)/\sqrt{d} \le \rho(d+1)$. Since $E^s h^{s-1}_* = 0$, boundedness and Lipschitz continuity give, simultaneously for every input,

$$e_s(x) \le (\sqrt{2}C^2 + \sqrt{C} + b_L)e_{s-1}(x) + \rho\sqrt{C}.$$

Put $A = 2 + 3C^2 + b_L$. Induction yields $e_s(x) \le \rho(d+L+1)A^{L-2}$ for $s < L$. At the output,

$$|z_w(x) - z_*(x)| \le 3\sqrt{2}C^2 e_{L-1}(x) + \rho.$$

Hence choosing

$$\rho = \left[2 + 9C^2(d+L+1)A^{L-2}\right]^{-1}$$

ensures output discrepancy at most $1/2$. All networks in the cylinder with the operator-norm event therefore implement f with margin greater than one.

It remains to price the cylinder. Its number of constrained coordinates is

$$m = N\big[d + 2 + (L-2)C\big].$$

The sum of their squared centers is at most NB_0, where one may take

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4.$$

Indeed, a first-layer center has magnitude at most $\sqrt{d}C$, an internal center at most $\sqrt{2C}C$, and a readout center at most $6C^2$. For $Z \sim N(0,t^2)$ and $|Z-a| \le \rho t$,

$$\Pr\{|Z-a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}}\exp\{-a^2/t^2 - \rho^2\}.$$

Multiplication and $t^{-2} = N^{\gamma-1}$ show

$$-\log\Pr(\mathcal{C}_\rho) \le N^\gamma B_0 + N[d+2+(L-2)C]\log\frac{e\sqrt{2\pi}}{2\rho}.$$

The desired probability is at least $\Pr(\mathcal{C}_\rho)/2$. Since $N \le N^\gamma$ and $N^\gamma \ge 1$, this proves the claim with

$$Q_L(d,C) = 2B_0 + \frac{4[d+2+(L-2)C]}{\rho} + 2.$$

Here we used $\log(e\sqrt{2\pi}/(2\rho)) \le 2/\rho$ for $0 < \rho \le 1$. The displayed expression is a polynomial in d, C at every fixed depth. □

3 An exact variational principle at the critical exponent

For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation. It provides the variational foundation for the compression argument that follows.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \dots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1, 1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E hh^{\mathsf{T}}\, d\nu(h), \qquad Q_0(a, b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = \big(\phi((Q^{1/2}u)_a)\big)_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho}\, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^+ is the Moore–Penrose inverse.

Theorem 3.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1, \dots, \nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D\big(\nu_s \| \mathsf{M}_{K(\nu_{s-1})}\big) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{1}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z \in O} I(z) \le \liminf_{N \to \infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N \to \infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z \in F} I(z).$$

The infimum in (1) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [1].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N}\sum_{i=1}^{N}\big(h_i^{s-1}(x^a)h_i^{s-1}(x^b)\big)_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2}V_N. \tag{2}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta\|\Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s\|\Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N}, \dots, \eta_{r,N}, V_N)$.

The recursive map from these variables to $(\nu_1, \dots, \nu_r, z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j}\, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q\, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (2).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta : T_{\#}\eta = \nu} D(\eta\|\rho) = D(\nu\|T_{\#}\rho). \tag{3}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_{\#}\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the identity in the extended real sense. Apply (3) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v : Q^{1/2}v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (1).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^{\mathsf{T}} z - \frac{1}{2} t^{\mathsf{T}} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (1) is attained. Moreover $K(\nu) \preceq mI_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary 3.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1, 1\}^m$, set*

$$J(f) = \inf_{z:\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N\to\infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f}) = J(f). \tag{4}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g\in\mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N\to\infty} -\frac{1}{N} \log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (1) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (4).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g\in\mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

The variational principle exhibits the distinction between the critical regime and a frozen kernel model. At $\gamma = 1$, deviations of a hidden feature law have an order-N entropy cost, at the same scale as an order-one output. Thus the optimization is over every hidden feature law, with a separate entropy charge for each layer. It is not in general legitimate to set those laws to their unconditioned values. The following sections obtain a polynomial description of the minimizing feature laws for tanh and use it to compress every Boolean function with bounded full-margin rate.

4 Quadratic tilts at a critical full-margin minimizer

We now specialize to $\phi = \tanh$. Its analyticity and its bounded first two derivatives yield additional structure at the minimizers of (1). The following argument uses the exact variational principle above; it does not assume mean-field equations.

Lemma 4.1 (Positive definite feature covariances)**.** *If $(\nu_1, \ldots, \nu_r, z)$ has finite cost in (1), then every $K(\nu_s)$ is positive definite.*

Proof. For any nonzero $c \in \mathbb{R}^m$, the function

$$H_c(b, v) = \sum_{a=1}^{m} c_a \tanh(b + v \cdot x^a)$$

is real analytic and is not identically zero. To see the latter, choose v_0 such that the numbers $v_0 \cdot x^a$ are all distinct. If H_c were identically zero, evaluating it at $(b, v) = (-t\theta, tv_0)$ and letting $t \to \infty$ would give

$$\sum_a c_a \operatorname{sign}(v_0 \cdot x^a - \theta) = 0$$

for every θ different from these numbers. The difference between two successive such identities is twice one coefficient c_a, so all coefficients would vanish. A nonzero real-analytic function has a Lebesgue-null zero set. The first-layer weights have a positive Gaussian density on the $(d+1)$-dimensional (b, v) space. Consequently

$$\mathsf{M}_{Q_0}\{h : c^{\mathsf{T}} h = 0\} = 0.$$

Finite entropy implies $\nu_1 \ll \mathsf{M}_{Q_0}$, and therefore $c^{\mathsf{T}} K(\nu_1) c > 0$ for every $c \neq 0$.

Inductively, if $Q = K(\nu_{s-1})$ is positive definite, then M_Q has a positive density on $(-1, 1)^m$, since coordinatewise tanh is a smooth bijection from $\mathbb{R}^m$ onto that set. Every proper linear hyperplane is M_Q-null. Absolute continuity $\nu_s \ll \mathsf{M}_Q$ again gives positive definiteness of $K(\nu_s)$. □

Theorem 4.2 (Quadratic form of the optimal feature laws)**.** *Let $a > 0$, $f \in \{-1, 1\}^m$, and suppose*

$$J_a(f) = \inf_{z:\ f_b z_b \geq a\ \forall b} I(z) \leq K < \infty.$$

There is a minimizing tuple $(\nu_1, \ldots, \nu_r, z)$. Write $Q_s = K(\nu_s)$ and $\beta = Q_r^{-1} z$. There are symmetric matrices $\Gamma_1, \ldots, \Gamma_r$ such that

$$\frac{d\nu_1}{d\mathsf{M}_{Q_0}}(h) = \frac{e^{h^{\mathsf{T}}\Gamma_1 h}}{Z_1}, \qquad \frac{d\nu_s}{d\mathsf{M}_{Q_{s-1}}}(h) = \frac{e^{h^{\mathsf{T}}\Gamma_s h}}{Z_s} \quad (2 \leq s \leq r). \tag{5}$$

Moreover,

$$\|\beta\|_1 \leq \frac{2K}{a}, \qquad \Gamma_r = \frac{1}{2}\beta\beta^{\mathsf{T}}. \tag{6}$$

If $g_s = \sum_{i,j} |(\Gamma_s)_{ij}|$, then

$$g_r \leq \frac{2K^2}{a^2}, \qquad g_s \leq 3g_{s+1} + 2g_{s+1}^2 \quad (1 \leq s < r). \tag{7}$$

Thus, at fixed depth, all these entrywise norms are bounded by a polynomial in K and $1/a$, independently of the number $m = 2^d$ of inputs.

Proof. Existence follows from the goodness of I, closedness of the margin constraints, and attainment of the feature-law infimum in Theorem 3.1. Every Q_s is positive definite by Lemma 4.1.

For fixed feature laws, the output minimizes the strictly convex quadratic $z^{\mathsf{T}} Q_r^{-1} z/2$ subject to the linear margin constraints. Their feasible set has a strictly feasible point. The elementary quadratic-program optimality conditions give multipliers $\lambda_b \geq 0$ such that

$$\beta_b = f_b \lambda_b, \qquad \lambda_b(f_b z_b - a) = 0.$$

It follows that

$$\frac{1}{2}\beta^{\mathsf{T}} z = \frac{a}{2}\sum_b \lambda_b = \frac{a}{2}\|\beta\|_1 \le J_a(f) \le K,$$

which proves the first bound in (6).

We justify the feature-law variations, including possible zeros of their densities. Fix a layer s, freeze every other variable, and denote its reference measure by μ_s (either M_{Q_0} or $\mathsf{M}_{Q_{s-1}}$). Its contribution to the objective has the form

$$D(\nu_s\|\mu_s) + F_s(K(\nu_s)) + \text{constant},$$

where F_s is differentiable near the current positive definite Q_s. For $s = r$, $F_r(Q) = z^{\mathsf{T}} Q^{-1} z/2$, so

$$\nabla F_r(Q_r) = -\frac{1}{2}\beta\beta^{\mathsf{T}}.$$

For $s < r$, put $\xi(h) = (\operatorname{arctanh}(h_b))_{b=1}^m$ and $B_{s+1} = \int \xi(h)\xi(h)^{\mathsf{T}}\, d\nu_{s+1}(h)$. This moment is finite: by invariance of relative entropy under the coordinatewise tanh bijection, the preactivation law has finite relative entropy with respect to $\mathcal{N}(0, Q_s)$, and the Gaussian exponential moment of a sufficiently small multiple of $\xi^{\mathsf{T}} Q_s^{-1}\xi$ controls its expectation by the entropy inequality. The Gaussian density ratio, whose Jacobian cancels under tanh, gives, near Q_s,

$$F_s(Q) = \frac{1}{2}\log\det Q + \frac{1}{2}\operatorname{tr}(Q^{-1}B_{s+1}) + \text{constant}.$$

In particular,

$$-\nabla F_s(Q_s) = \frac{1}{2}Q_s^{-1}(B_{s+1} - Q_s)Q_s^{-1} =: \Gamma_s. \tag{8}$$

Put $B = \nabla F_s(Q_s)$ and let ρ_s have density $e^{-h^{\mathsf{T}}Bh}/Z$ with respect to μ_s. Its logarithm is bounded, so $D(\nu_s\|\rho_s)$ is finite. For $\nu_{s,t} = (1-t)\nu_s + t\rho_s$, convexity of relative entropy gives

$$D(\nu_{s,t}\|\rho_s) - D(\nu_s\|\rho_s) \le -tD(\nu_s\|\rho_s).$$

The identity $D(\nu\|\rho_s) = D(\nu\|\mu_s) + \int h^{\mathsf{T}}Bh\, d\nu + \log Z$ and differentiability of F_s imply that the full objective changes by at most $-tD(\nu_s\|\rho_s) + o(t)$. Minimality forces $D(\nu_s\|\rho_s) = 0$. Thus $\nu_s = \rho_s$, proving (5) with $\Gamma_s = -\nabla F_s(Q_s)$. This also proves $\Gamma_r = \beta\beta^{\mathsf{T}}/2$.

It remains to bound the apparently ill-conditioned expression (8) without using $\|Q_s^{-1}\|$. Under ν_{s+1} the preactivation vector ξ has Gaussian reference $\mathcal{N}(0, Q_s)$ tilted by $e^{V(\xi)}$, where

$$V(\xi) = \tanh(\xi)^{\mathsf{T}}\Gamma_{s+1}\tanh(\xi).$$

Twice applying Gaussian integration by parts gives

$$B_{s+1} - Q_s = Q_s\,\mathbb{E}_{\text{tilt}}[\nabla^2 V + (\nabla V)(\nabla V)^{\mathsf{T}}]\,Q_s.$$

This identity is justified directly because V and its first two derivatives are bounded. Substitution into (8) gives

$$\Gamma_s = \frac{1}{2}\mathbb{E}_{\text{tilt}}[\nabla^2 V + (\nabla V)(\nabla V)^{\mathsf{T}}].$$

The bounds $|\tanh| \le 1$, $|\tanh'| \le 1$, and $|\tanh''| \le 2$ imply

$$\|\nabla V\|_1 \le 2g_{s+1}, \qquad \sum_{i,j}|\partial_i\partial_j V| \le 6g_{s+1}.$$

Therefore $g_s \le 3g_{s+1} + 2g_{s+1}^2$. The final-layer bound is $g_r = \|\beta\|_1^2/2 \le 2K^2/a^2$. Iterating the displayed quadratic recursion a fixed number of times proves the asserted polynomial dependence. □

5 Compression of bounded quadratic-tilt towers

This section uses $\phi = \tanh$. Write $\|A\|_{\max} = \max_{ij} |A_{ij}|$ and $\|A\|_{1,\mathrm{ent}} = \sum_{ij} |A_{ij}|$ for matrices on the finite input set. These are different norms; their dual pairing is what avoids a factor of 2^d below. For a positive semidefinite covariance Q and a symmetric matrix Γ, let

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\phi(Z)\phi(Z)^\top \exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}.$$

All activations are coordinatewise. The definition is valid for singular Q.

Lemma 5.1 (A dimension-independent covariance comparison)**.** *If $g = \|\Gamma\|_{1,\mathrm{ent}}$, then for all positive semidefinite Q, Q' of the same size,*

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1+g)^2 \|Q - Q'\|_{\max}.$$

Proof. Put $V(z) = \phi(z)^\top \Gamma \phi(z)$ and $Q_t = (1-t)Q + tQ'$. For a twice differentiable bounded test function F, let $\mathbb{E}_t$ denote expectation under the Gaussian of covariance Q_t reweighted by e^V. Gaussian differentiation and the quotient rule give

$$\frac{d}{dt}\mathbb{E}_t F = \frac{1}{2}\sum_{a,b} (Q'-Q)_{ab} \{\mathbb{E}_t[\partial_{ab}F + \partial_a F \partial_b V + \partial_b F \partial_a V] + \mathrm{Cov}_t(F, \partial_{ab}V + \partial_a V \partial_b V)\}.$$

This formula has no reciprocal partition-function bound: all terms are expectations under a probability measure. Define $M_1(H) = \sup_z \sum_a |\partial_a H(z)|$ and $M_2(H) = \sup_z \sum_{a,b} |\partial_{ab} H(z)|$. Because $|\phi|, |\phi'| \le 1$ and $|\phi''| \le 2$,

$$M_1(V) \le 2g, \qquad M_2(V) \le 6g.$$

For $F(z) = \phi(z_i)\phi(z_j)$, including $i = j$, $\|F\|_\infty \le 1$, $M_1(F) \le 2$, and $M_2(F) \le 6$. The absolute derivative is therefore at most

$$\frac{1}{2}\|Q - Q'\|_{\max}\{6 + 8g + 2(6g + 4g^2)\} = (3 + 10g + 4g^2)\|Q - Q'\|_{\max}.$$

Integrating in t proves the assertion, with the displayed slightly larger constant. One may initially replace both covariances by $Q + \epsilon I$ and $Q' + \epsilon I$. The functions and all derivatives used above are bounded, and the partition functions lie between e^{-g} and e^g, so weak Gaussian convergence lets $\epsilon \downarrow 0$. This also proves the singular case. □

Proposition 5.2 (Finite realization of a bounded-tilt tower)**.** *Fix $L \ge 2$, $K \ge 1$, and $G \ge 1$. Let Q_0 be the covariance of the Gaussian first-layer field $w \cdot \bar{x}$, with $w \sim \mathcal{N}(0, I_{d+1}/d)$, indexed by $x \in \mathcal{X}$. Suppose symmetric matrices $\Gamma_1, \dots, \Gamma_{L-1}$ and a vector $\beta \in \mathbb{R}^{\mathcal{X}}$ satisfy*

$$\|\Gamma_s\|_{1,\mathrm{ent}} \le G, \qquad \|\beta\|_1 \le 4K, \qquad Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1}).$$

Then the function $z(x) = (Q_{L-1}\beta)_x$ can be uniformly approximated to accuracy $1/8$ by a depth-L network whose width and weights are bounded by $\mathrm{poly}_L(d, K, G)$. If $f(x)z(x) \ge 1/2$ everywhere, a factor-two increase in the finite network's readout implements f with margin at least $3/4$.

Proof. Let $b = 8(1+G)^2$ and

$$\eta = [32K(L-1)b^{L-2}]^{-1}.$$

Choose the common width n to be the least integer satisfying

$$n \geq 2\eta^{-2}\log(8(L-1)\,2^{2d}).$$

The construction is random only to prove existence. The matrices Γ_s may be indexed by the entire cube; they select the sampling laws, but are not stored or evaluated by the resulting finite network. No efficient sampling or construction algorithm is claimed. At the first layer, independently sample n Gaussian weight vectors $w_i \sim \mathcal{N}(0, I_{d+1}/d)$ tilted by

$$\exp\{h(w_i)^\top \Gamma_1 h(w_i)\}, \qquad h(w_i)_x = \phi(w_i \cdot \bar{x}).$$

Their activation vectors have exactly the law defining $Q_1 = \mathsf{F}_{\Gamma_1}(Q_0)$. At each subsequent hidden layer, conditional on the already sampled previous activation vectors $h_1, \ldots, h_n$, independently sample n standard Gaussian row vectors $\xi_j \in \mathbb{R}^n$ tilted by

$$\exp\{\phi(Z_j)^\top \Gamma_s \phi(Z_j)\}, \qquad Z_j(x) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \xi_{ji} h_i(x).$$

Use the actual network weights $\xi_{ji}/\sqrt{n}$. If $\widehat{Q}_{s-1} = n^{-1}\sum_i h_i h_i^\top$ is the preceding empirical activation covariance, then, conditionally, Z_j before tilting is Gaussian with exactly that covariance. Thus the new activation vectors are conditionally independent, with second moment $\mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})$.

Every covariance entry is the empirical mean of n variables in $[-1,1]$. Hoeffding's inequality, a union bound over 2^{2d} entries, and conditioning at successive layers show, with probability at least $3/4$, that

$$\|\widehat{Q}_s - \mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})\|_{\max} \leq \eta \quad (1 \leq s < L),$$

where $\widehat{Q}_0 = Q_0$. Lemma 5.1 then implies

$$\|\widehat{Q}_{L-1} - Q_{L-1}\|_{\max} \leq (L-1)b^{L-2}\eta.$$

For each last-layer activation vector h_i, choose the deterministic readout weight $v_i = (\beta \cdot h_i)/n$. The resulting output is $\widehat{z}(x) = (\widehat{Q}_{L-1}\beta)_x$, and therefore

$$\|\widehat{z} - z\|_\infty \leq 4K(L-1)b^{L-2}\eta = 1/8.$$

Although β is indexed by the entire input cube, its only role in the finite network is to select these n real readout weights; each satisfies $|v_i| \leq 4K/n$. It is not an additional computational input to the network.

It remains to bound the sampled weights without losing polynomial size. For every sampled row, its tilted density relative to its Gaussian reference is at most e^{2G}, since the quadratic tilt lies in $[-G, G]$. Let $D_n = n(d+1) + (L-2)n^2$ be the total number of incoming scalar coordinates, and choose

$$B = \left[4G + 2\log(8\max\{D_n, 1\})\right]^{1/2}.$$

Each raw incoming coordinate has reference variance at most one, so, also conditionally on all preceding layers, its probability of exceeding B in absolute value is at most $2e^{2G - B^2/2}$. A union bound gives probability at most $1/4$ that any incoming coordinate exceeds B. The actual internal weights divide the raw coordinates by $\sqrt{n}$ and so satisfy the same bound. Intersecting this event with the covariance-approximation event leaves probability at least $1/2$. Fix a realization in the intersection. Both n and $\max\{B, 8K/n\}$ after the factor-two readout increase are bounded by polynomials in d, K, G at fixed depth. The stated margin follows from $2(1/2 - 1/8) = 3/4$. □

Lemma 5.3 (Counting finite realizations)**.** *Fix L, width n, and an entrywise bound $B \geq 1$. The Boolean functions having depth-L, width-n implementations with margin at least $3/4$ and weights bounded by B number at most*

$$\exp\{P \log(3 + 16B\Lambda)\}, \qquad P = n(d+1) + (L-2)n^2 + n, \quad \Lambda = (d + Ln)[1 + n(B+1)]^{L-1}.$$

Proof. Round every weight to a multiple of $(8\Lambda)^{-1}$. The first hidden layer's coordinatewise error is at most $(d+1)/(8\Lambda)$. Every subsequent hidden layer has error at most $n(B+1)$ times the preceding error plus $n/(8\Lambda)$; the same estimate applies at the linear readout. Induction bounds the final error by $1/8$. Each rounded coordinate has at most $3 + 16B\Lambda$ possible values, so the P-parameter grid has the stated size. Rounding preserves every represented Boolean sign function. □

6 Critical rate complexity and Bayesian generalization

Theorem 6.1 (Polynomial rate-sublevel counting)**.** *For every fixed depth $L \geq 2$, with activation* tanh*, there is a polynomial B_L such that, for every $K \geq 1$,*

$$\log \#\{f : \mathcal{X} \to \{-1, 1\} : J(f) \leq K\} \leq B_L(d, K).$$

Moreover every function in this sublevel set has a depth-L implementation of width and weight magnitude at most $\mathrm{poly}_L(d, K)$*.*

Proof. Take a minimizing tuple for $J(f) \leq K$ in Theorem 4.2, with margin $a = 1/2$. Its vector β satisfies $\|\beta\|_1 \leq 4K$. Let $g_r^* = 8K^2$ and recursively $g_s^* = 3g_{s+1}^* + 2(g_{s+1}^*)^2$ for $s < r = L - 1$. Then $G = 1 + \max_s g_s^*$ is a polynomial in K at fixed depth, and all the quadratic tilt matrices have entrywise norm at most G. The feature covariances satisfy $Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1})$, and the optimal output is $z = Q_r\beta$. Proposition 5.2 supplies a finite implementation with margin at least $3/4$, common width bounded by a polynomial in d, K, and weights bounded by another such polynomial. The bounds in its construction can be chosen uniformly for every function in the sublevel set. Lemma 5.3 counts the resulting fixed parameter grid and gives the stated bound. For the complexity statement, increase width by zero padding if necessary so that the width also bounds every weight magnitude. This preserves the network for tanh, since $\tanh(0) = 0$. □

Proof of Theorem 1.1. The uniform cloning bound and the exact margin rate imply

$$J(f) \leq Q_L(d, C)/2, \qquad C = C_L(f).$$

Let

$$\mathcal{A}_f = \{g : J(g) \leq J(f) + 1\}.$$

Its logarithmic cardinality is bounded by $B_L(d, Q_L(d, C)/2 + 1)$, and hence by a polynomial $p_L(d, C)$. For each $g \notin \mathcal{A}_f$, the exact rate formula gives

$$\frac{\mu_{N,1}(A_{N,g})}{\mu_{N,1}(A_{N,f})} \longrightarrow 0 \quad (N \to \infty).$$

Indeed the rate gap is greater than one; an infinite rate causes no problem. At fixed d there are finitely many Boolean functions, so summing these ratios yields

$$\frac{\pi_{N,1}(\mathcal{A}_f^c)}{\pi_{N,1}(f)} \longrightarrow 0.$$

Thus there is $N_0 \geq C$ such that for every $N \geq N_0$ this ratio is at most $\varepsilon/6$. This width threshold depends on the target and accuracy, but not on the input distribution or training sample. Lemma 8.2, with $a = \tau = \varepsilon/6$ and $H = p_L(d, C)$, now gives $R_D(\Pi_{N,1}) \leq \varepsilon/3$ with probability at least $1 - \delta$ whenever

$$n \geq \frac{6}{\varepsilon}[p_L(d, C) + \log(1/\delta)].$$

At every input where the correct-label posterior probability is at most $2/3$, the posterior error is at least $1/3$. The distributional mass of such inputs is at most $3R_D(\Pi_{N,1}) \leq \varepsilon$. This proves the claimed confidence guarantee, including its strict success threshold. □

7 Shallow norm compression and counting

Lemma 7.1 (Explicit shallow counting)**.** *For $T \geq 1$, let $\mathcal{A}_{N,T}$ be the set of Boolean functions implemented with strict signed margin greater than $1/2$ at every input and $\kappa_N(w) \leq T$ by a width-N network with $L = 2$. Let*

$$m = \lceil 512(T+1)((d+1)\log 2 + 1) \rceil, \quad H(d, T) = m(d+2)\log(2^{14}m(d+1)(T+1)^2).$$

Then $\log |\mathcal{A}_{N,T}| \leq H(d, T)$, independently of N.

Proof. Write the incoming rows as $u_i \in \mathbb{R}^{d+1}$ and the readout entries as v_i. The norm bound implies

$$\sum_i \|u_i\|_2^2 \leq NT/d, \quad \sum_i v_i^2 \leq T/N, \quad \sum_i |v_i| \leq \sqrt{T}.$$

Delete terms with $\|u_i\|_2 > B$, where $B = 16T/\sqrt{d}$. There are at most $NT/(dB^2)$ such terms. Cauchy–Schwarz bounds their uniform contribution to the output by $T/(\sqrt{d}B) = 1/16$. For the retained terms put $S = \sum_i |v_i| \leq \sqrt{T}$. An implementing network cannot have $S = 0$ after this deletion, since its original absolute output was everywhere greater than $1/2$. Draw m retained indices independently with probabilities $|v_i|/S$. The empirical network

$$\widetilde{z}(x) = \frac{S}{m}\sum_{j=1}^{m} \operatorname{sgn}(v_{i_j})\phi(u_{i_j} \cdot \bar{x})$$

has expectation equal to the retained network. For each input, Hoeffding's inequality gives

$$\mathbb{P}\{|\widetilde{z}(x) - \mathbb{E}\widetilde{z}(x)| > 1/16\} \leq 2\exp[-m/(512T)].$$

A union bound over the 2^d inputs makes the failure probability less than one for the stated m. Fix a realization with uniform error at most $1/16$. The total error from the original network is at most $1/8$. Doubling the readout now gives a width-m implementation with margin strictly greater than $3/4$, incoming coordinates bounded by $16T$, and readout absolute sum at most $2\sqrt{T}$.

Round each incoming coordinate to the nearest multiple of $\eta_1 = [64\sqrt{T+1}(d+1)]^{-1}$ and each readout coordinate to the nearest multiple of $\eta_2 = (32m)^{-1}$. Rounding incoming weights changes the output by at most $2\sqrt{T}(d+1)\eta_1 \leq 1/32$; rounding the readout changes it by at most $m\eta_2 = 1/32$. These conservative bounds allow rounding errors of a whole mesh step. The resulting grid network preserves the sign of the implementation at every input. Each incoming coordinate has at most $3 + 2048(d+1)(T+1)^{3/2}$ grid choices. Each readout coordinate, bounded before rounding by $2\sqrt{T}$, has at most $3 + 128m\sqrt{T+1}$ grid choices. Their product, over $m(d+1)$ incoming and m readout coordinates, is at most $\exp H(d, T)$. Every function in the class has a sign-preserving representative on this one fixed grid. A grid network computes at most one Boolean sign function, so the same bound counts the functions themselves. □

8 The posterior bound

Lemma 8.1 (Gaussian tail)**.** *If $X \sim \chi^2_D$ and $u \geq 8D$, then $\mathbb{P}(X \geq u) \leq e^{-u/4}$.*

Proof. Chernoff's inequality at parameter $3/8$ gives $\mathbb{P}(X \geq u) \leq 4^{D/2} e^{-3u/8}$. Since $D \leq u/8$, its logarithm is at most $u \log 2/8 - 3u/8 \leq -u/4$. □

Lemma 8.2 (Finite relevant class)**.** *Let a prior on a finite class assign positive mass to a target f. Suppose $f \in \mathcal{A}$, $|\mathcal{A}| \leq e^H$, and $\pi(\mathcal{A}^c) \leq \tau\pi(f)$. For any $a, \delta \in (0,1)$, if $n \geq a^{-1}(H + \log(1/\delta))$, then with probability at least $1-\delta$ its consistency posterior satisfies $R_D(\Pi) \leq a + \tau$.*

Proof. Any fixed function with error greater than a survives n labels with probability at most $(1-a)^n \leq e^{-an}$. A union bound excludes all such survivors in $\mathcal{A}$ except on an event of probability e^{H-an}. For every sample, the posterior normalizing probability is at least $\pi(f)$, so its mass outside $\mathcal{A}$ is at most τ. □

Proof of Theorem 1.2. Let $Q = Q_2(d, C)$ be the explicit cloning bound above, so $\mu_{N,\gamma}(A_{N,f}) \geq e^{-N^\gamma Q/2}$. Choose

$$T = \max\{1,\ 8(d+2),\ 2Q + 4\log(2/\varepsilon)\}.$$

At $L = 2$ the number of Gaussian parameters is $D_0 = N(d+2)$ and $N^\gamma \kappa_N(w) \sim \chi^2_{D_0}$. Since $\gamma \geq 1$, $D_0 \leq (d+2)N^\gamma$, so the preceding Gaussian bound yields

$$\frac{\mu_{N,\gamma}\{\kappa_N > T\}}{\mu_{N,\gamma}(A_{N,f})} \leq \exp[-N^\gamma(T/4 - Q/2)] \leq \varepsilon/2.$$

In particular $A_{N,f}$ contains a point with norm at most T, and thus $f \in \mathcal{A}_{N,T}$. If a function is outside $\mathcal{A}_{N,T}$, every one of its implementing weights has norm greater than T. Disjointness of the implementing events and cancellation of the global-margin normalization therefore give

$$\frac{\pi_{N,\gamma}(\mathcal{A}^c_{N,T})}{\pi_{N,\gamma}(f)} \leq \varepsilon/2.$$

Apply Lemmas 7.1 and 8.2 with $a = \tau = \varepsilon/2$ and $H = H(d,T)$. The explicit quantity $H(d,T)$ is bounded by a polynomial in $d, C, \log(2/\varepsilon)$, proving the theorem. For posterior majority, every misclassified input has posterior probability of error at least $1/2$; integration gives its error at most $2R_D(\Pi)$. □

9 Scope and relation to earlier mean-field arguments

At $\gamma = 1$ and depth $L \geq 3$, the typical weighted squared norm is of order N because there are $(L-2)N^2$ internal weights. The shallow norm-tail argument therefore cannot be used directly. Our all-depth proof instead bounds the coefficients of the optimal feature-law tilts and samples finite networks from their tilted Gaussian rows. The strong-prior parameter γ in the common cloning construction does not enter that endpoint variational argument.

The all-depth theorem is asymptotic in width, at fixed input dimension and depth. Its sample polynomial has width-independent coefficients, while its width threshold follows from a finite union of large-deviation comparisons and is not quantified. The shallow theorem holds at every width $N \geq C$. For larger depth the arguments in this paper do not establish an all-width endpoint theorem, a polynomial bound on the width threshold, or the same all-depth conclusion for every

bounded Lipschitz activation. Tanh supplies analyticity, invertibility, and bounded first and second derivatives at specific steps of our proof.

Vaintrob and Hänni [2] propose an all-depth mean-field compression argument using a finite-dimensional active subspace and fresh Gaussian fluctuations in its complement. There are two independent reasons to supply a separate argument for the all-depth conclusion here. First, its learning rule uses a smooth mean-squared-error Gibbs tilt, and its architecture has independent biases at every hidden layer. Our prior instead has a global Boolean margin restriction followed by exact label conditioning. Second, the displayed regularity invariant used in its proof is false as stated. We give a direct check, rather than assume that a change of conventions would repair it.

Proposition 9.1 (Failure of the stated pointwise sensitivity invariant)**.** *Let μ be uniform on $\mathcal{X}$, $M \geq 1$, and $S(\xi)(x) = M^{-1} \sum_{j=1}^{M} \tanh(\xi_j(x))$. If the bound*

$$|D_{j,u} S(\xi)(x)| \leq \frac{A}{\sqrt{M}} \|u\|_{L^2(\mu)} \tag{$*$}$$

holds for every field ξ, input x, channel j, and direction $u \in L^2(\mu)$, then $A \geq 2^{d/2}/\sqrt{M}$.

Proof. Set $\xi = 0$ and $u = 2^{d/2} \mathbf{1}_{\{x\}}$. Then $\|u\|_{L^2(\mu)} = 1$, whereas $D_{j,u} S(0)(x) = 2^{d/2}/M$. Substitution into $(*)$ gives the result. □

The invariant in Appendix A, Definition A.6(I4), of [2] has precisely the form $(*)$. Appendix B, Theorem B.14, asserts its initialization for the readout suffix with polynomial width and polynomial regularity constant. The proposition contradicts that assertion for polynomial M as d grows. A pointwise derivative estimate involving $|u(x)|$, or an averaged estimate for population loss, can be true. Replacing $(*)$ by either one changes what has to be proved in the subsequent induction, particularly for the empirical loss and conditional lazy-coordinate scores.

This identifies a gap in that earlier argument, not a counterexample to mean-field learnability. The present proof does not use the false sensitivity invariant, a conditional lazy-score bound, or a propagation-of-chaos premise. Its exact finite-width distributional representation and Gaussian covariance comparison yield the required count directly.

References

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